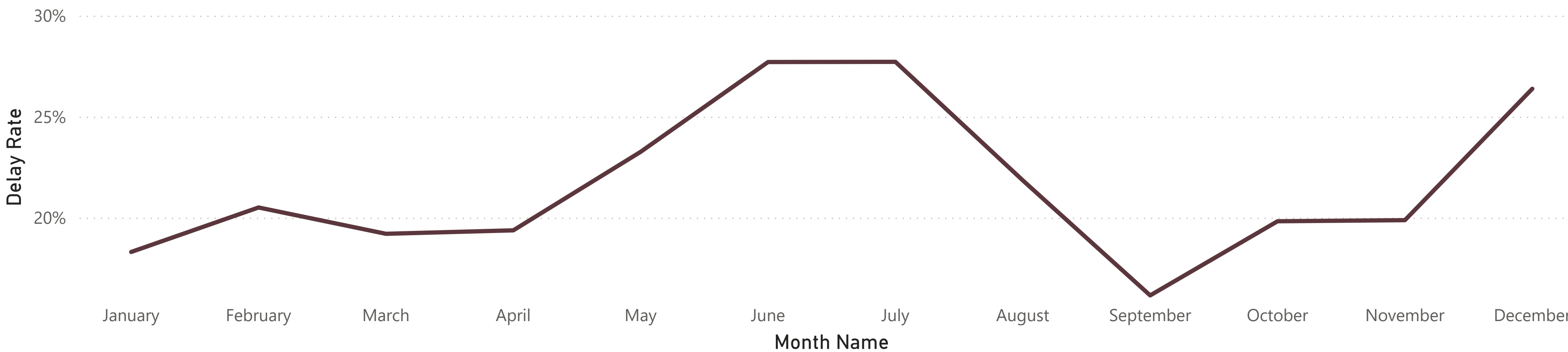
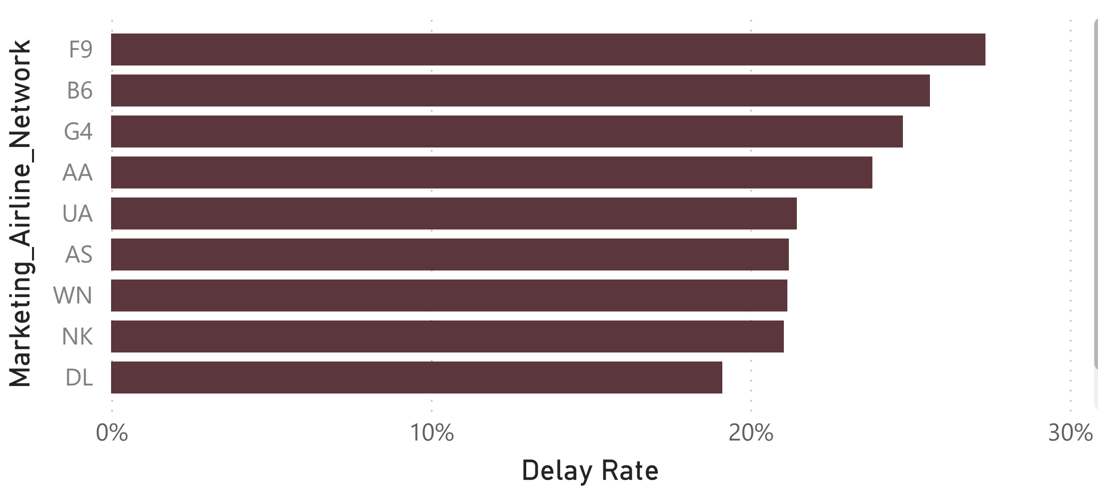
8M	76.42%	21.78%	1.53%	17.4	132.14M
Total Flights	On-Time Rate	Delay Rate	Cancellation Rate	Average Arrival Delay	Total Delay Minutes

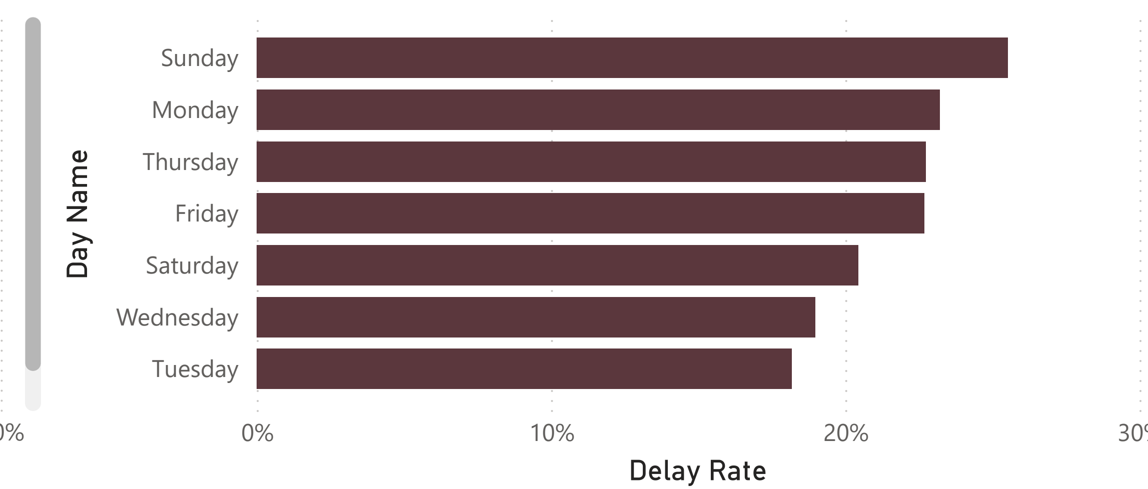
Delay Rate by Month



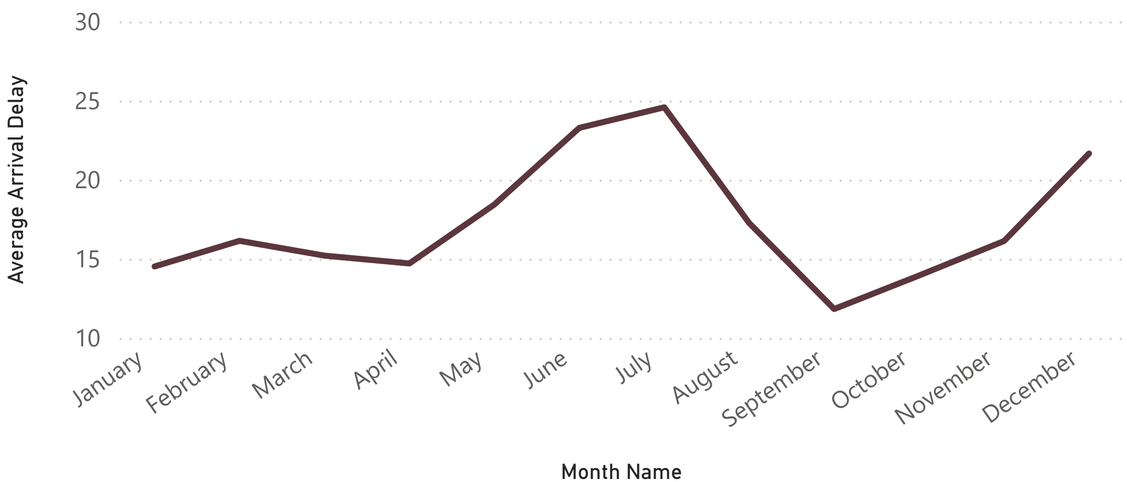
Delay Rate by Marketing Airline Network



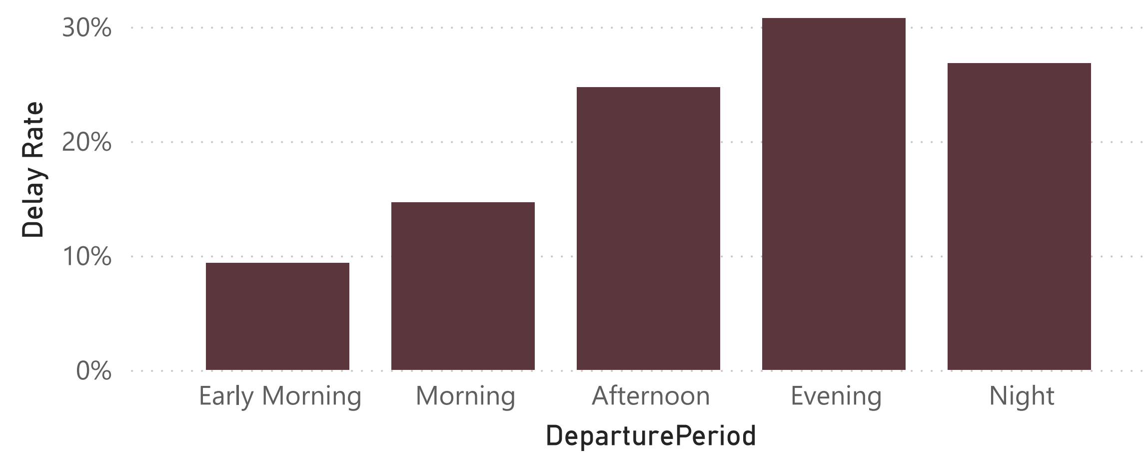
Delay Rate by Day



Average Arrival Delay by Month Name

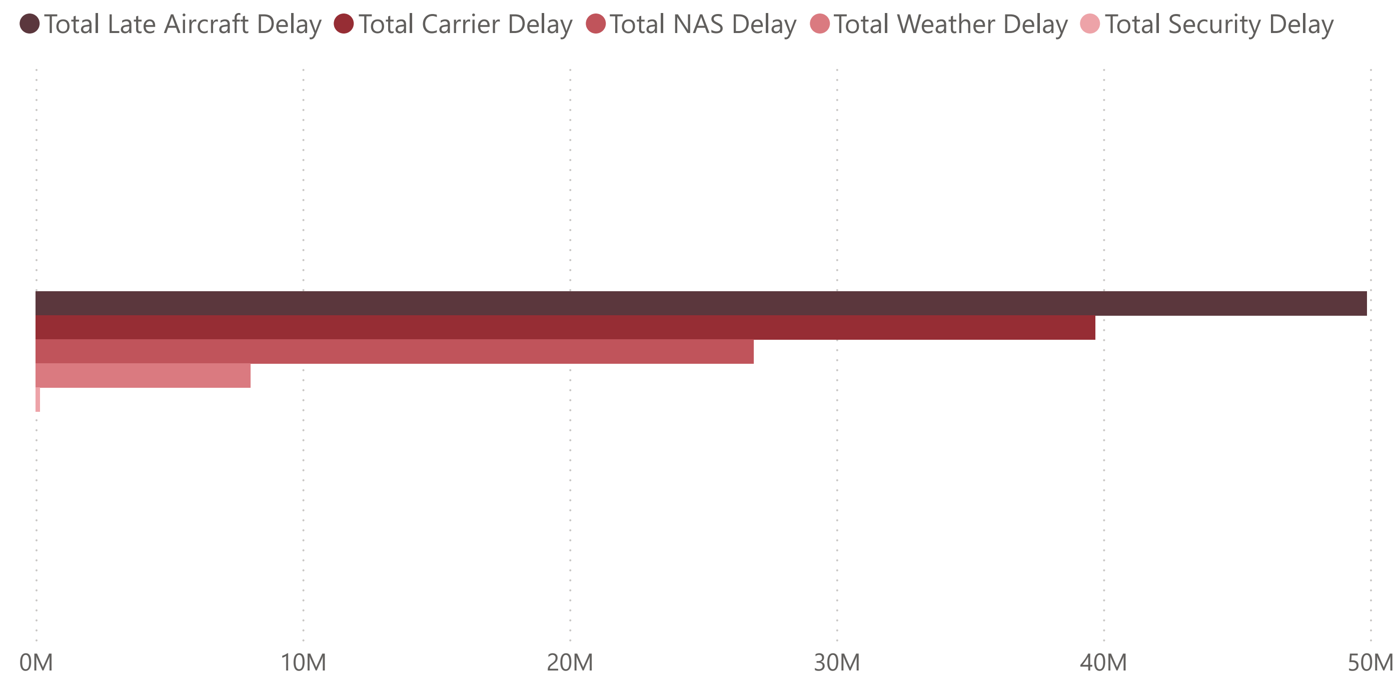


Delay Rate by DeparturePeriod

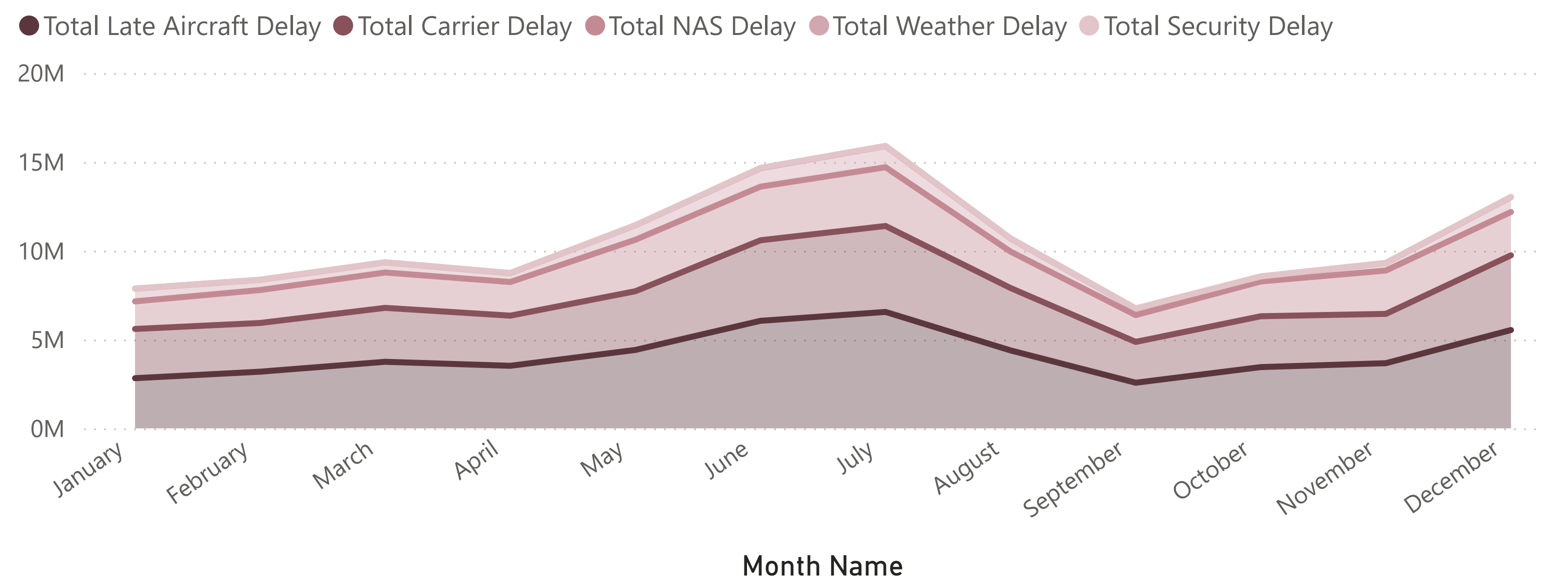


Month Name	Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	**Total**
January	21.69%	22.25%	15.09%	13.29%	16.76%	18.84%	20.99%	**18.30%**
February	22.00%	20.49%	16.04%	20.21%	28.07%	17.46%	18.25%	**20.50%**
March	25.47%	22.42%	12.64%	16.10%	17.15%	19.01%	18.59%	**19.20%**
April	19.52%	18.58%	13.75%	19.00%	23.58%	23.86%	17.97%	**19.37%**
May	25.79%	27.06%	21.84%	20.81%	22.73%	25.44%	18.77%	**23.26%**
June	30.23%	25.99%	25.30%	26.31%	30.01%	29.94%	25.39%	**27.70%**
July	30.49%	29.82%	25.90%	26.69%	29.58%	26.58%	24.49%	**27.71%**
August	24.23%	21.69%	16.29%	19.28%	22.68%	24.61%	21.94%	**21.84%**
September	17.42%	15.57%	12.24%	14.46%	20.60%	18.05%	14.71%	**16.15%**
October	25.26%	24.68%	16.77%	14.41%	19.59%	20.23%	17.61%	**19.82%**
November	27.45%	22.50%	16.11%	14.48%	18.11%	17.59%	19.87%	**19.87%**
December	34.39%	27.13%	23.62%	20.96%	23.00%	29.86%	26.08%	**26.38%**
Total	**25.52%**	**23.22%**	**18.19%**	**18.97%**	**22.74%**	**22.68%**	**20.45%**	**21.78%**

Total Delay Minutes by Cause



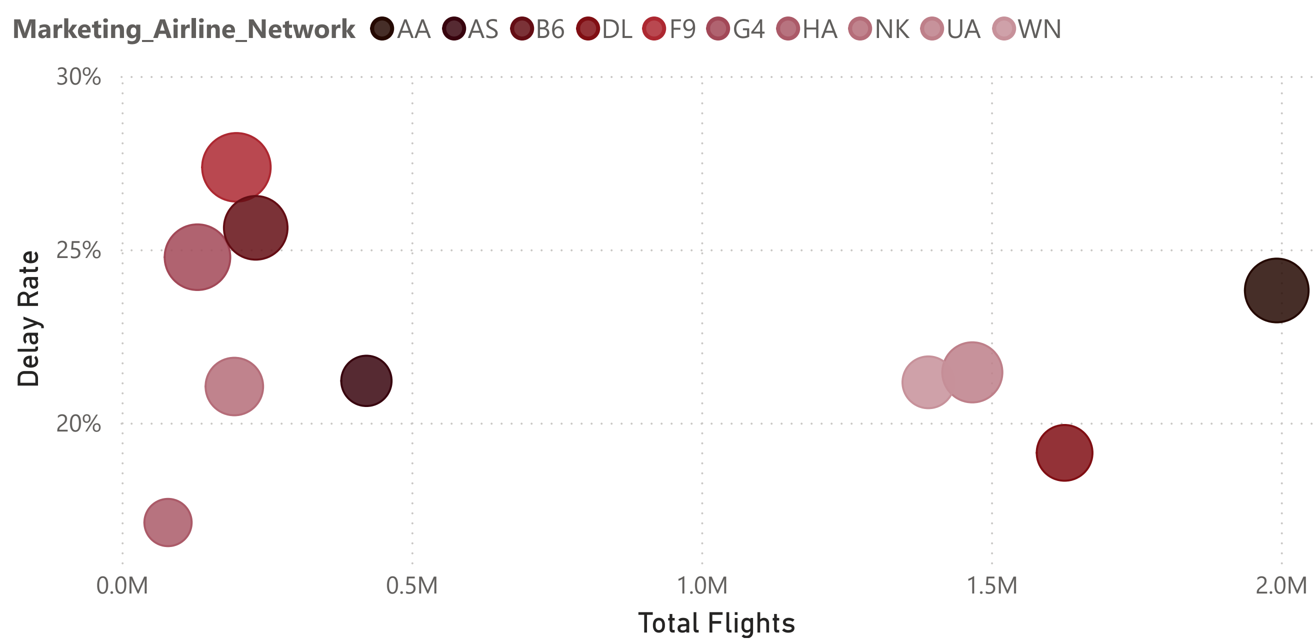
Cause Composition Over Time



Airline Ranking Table

Marketing_Airline_Network	Total Flights	Delay Rate	Average Arrival Delay	Cancellation Rate
F9	198065	27.37%	24.5	1.77%
B6	231413	25.62%	20.8	1.65%
G4	130899	24.77%	22.3	0.47%
AA	1992906	23.81%	20.9	2.36%
UA	1467730	21.45%	18.4	1.36%
AS	422375	21.21%	12.5	1.30%
WN	1391885	21.16%	13.3	0.85%
NK	194515	21.04%	16.8	1.50%
DL	1626898	19.13%	15.7	1.36%
HA	80084	17.12%	10.9	0.82%
Total	**7736770**	**21.78%**	**17.4**	**1.53%**

Flight Volume vs Delay Rate scatter plot



Analysis of 7.7M U.S. domestic flights in 2025 using Bureau of Transportation Statistics data. Findings describe observed patterns and associations in the data and should not be interpreted as confirmed causal relationships.

Business Implication	Evidence	Finding	Recommendation
Delay performance deteriorates sharply during the summer peak rather than remaining stable throughout the year.	July reached 28.59% delayed flights and 24.6 min average arrival delay, while September fell to 16.26% and 11.8 min.	Summer operations show a clear delay surge	Review June–July capacity, scheduling and operational buffers before peak season.
Even though F9 has a higher delay rate, AA's scale means its operational issues affect a much larger absolute volume of flights.	AA operated about 1.99M flights, the largest volume in the dataset, with a 23.81% delay rate.	AA combines high delay exposure with exceptional operating scale	Prioritize high-volume carriers for operational review using both delay rate and absolute delay impact.
Internal operational factors represent a much larger recorded delay burden than weather-related causes.	Late Aircraft Delay = 40.01% and Carrier Delay = 31.84%, together accounting for 71.85% of recorded delay-cause minutes. Weather accounted for only 6.45%.	Delay burden is concentrated in aircraft and carrier operations	Prioritize turnaround, aircraft rotation and carrier-side scheduling improvements.
The pattern is consistent with delays accumulating or propagating throughout the operating day.	Delay rate rises from 9.33% for early-morning departures to 30.73% for evening departures, more than 3× higher.	Delays increase sharply as the operating day progresses	Investigate turnaround time, aircraft rotation and schedule recovery capacity during later operating periods.